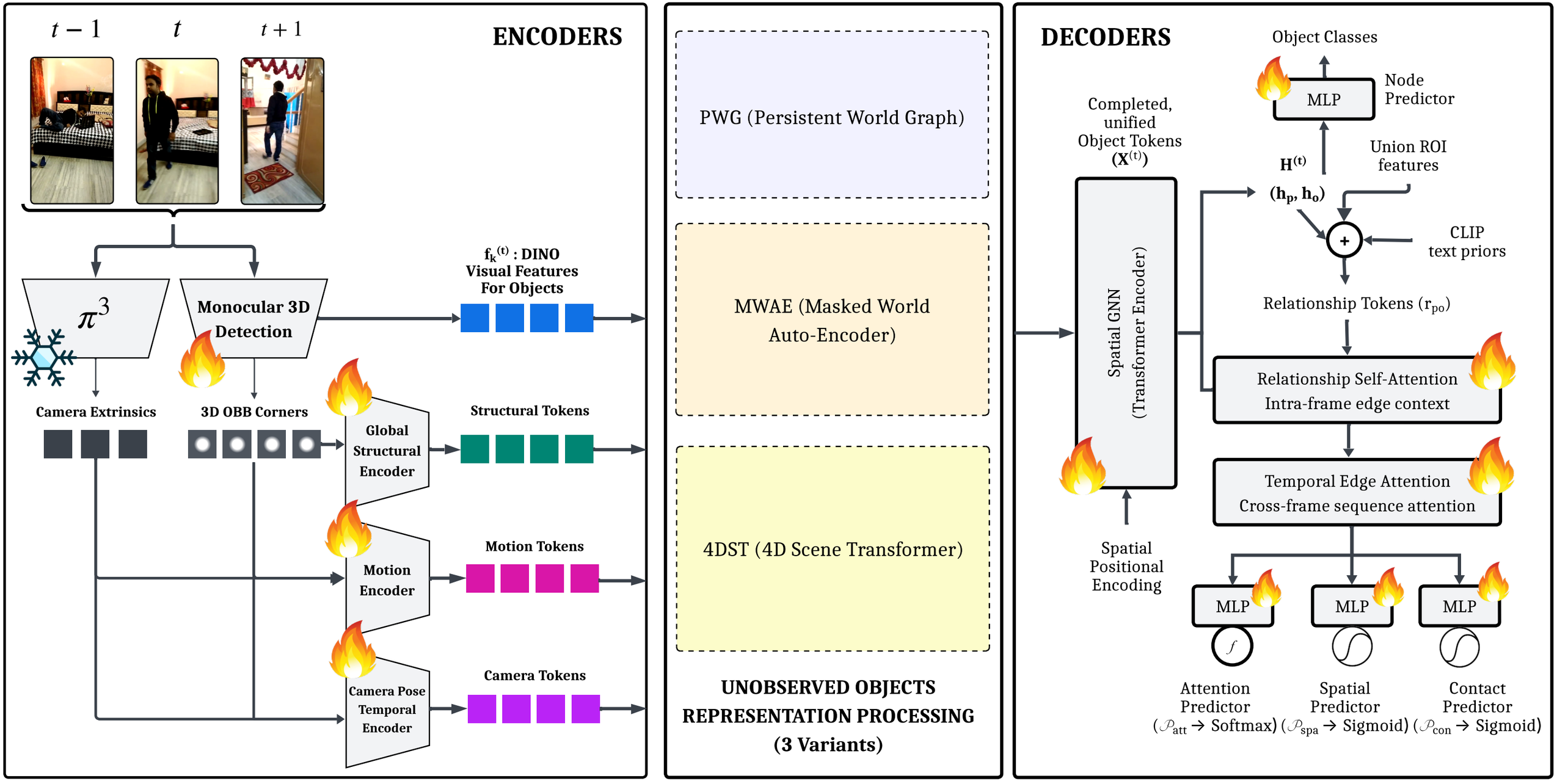
ENCODERS
$t-1$
t
$t+1$
π^3
Monocular 3D Detection
$f_k^{(t)}$: DINO Visual Features For Objects
Camera Extrinsics
3D OBB Corners
Global Structural Encoder
Structural Tokens
Motion Encoder
Motion Tokens
Camera Pose Temporal Encoder
Camera Tokens
PWG (Persistent World Graph)
MWAE (Masked World Auto-Encoder)
4DST (4D Scene Transformer)
UNOBSERVED OBJECTS REPRESENTATION PROCESSING (3 Variants)
DECODERS
Completed, unified Object Tokens ($X^{(t)}$)
Spatial GNN (Transformer Encoder)
Spatial Positional Encoding
$H^{(t)}$
(h_p, h_o)
Object Classes
MLP
Node Predictor
Union ROI features
CLIP text priors
Relationship Tokens (r_{po})
Relationship Self-Attention
Intra-frame edge context
Temporal Edge Attention
Cross-frame sequence attention
MLP
MLP
MLP
Attention Predictor ($\mathcal{P}_{att}$ → Softmax)
Spatial Predictor ($\mathcal{P}_{spa}$ → Sigmoid)
Contact Predictor ($\mathcal{P}_{con}$ → Sigmoid)